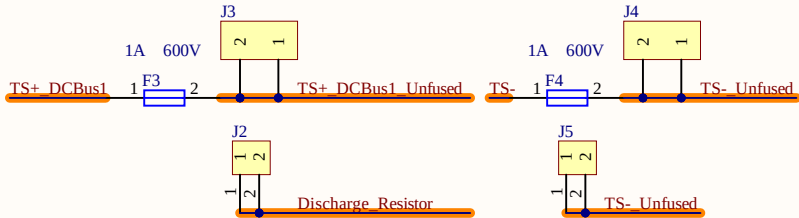
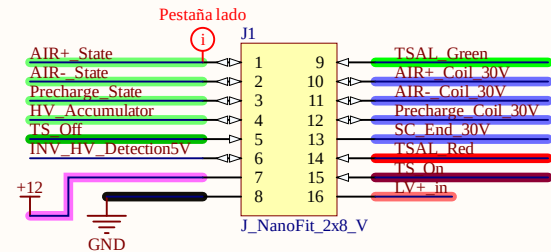


This PCB is installed in the power distribution box (HV Box) and is capable of controlling TSAL_Light according to 2025 FSG rules. It needs signals from AMS_Master (or TSAL_Dummy) and Voltage measurements on the DC Bus.

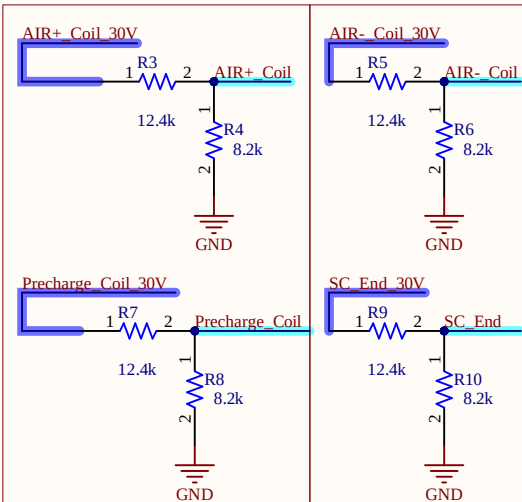
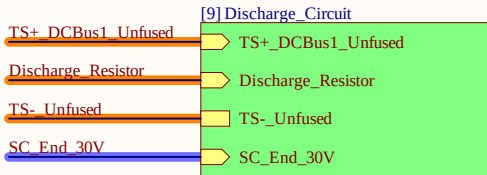
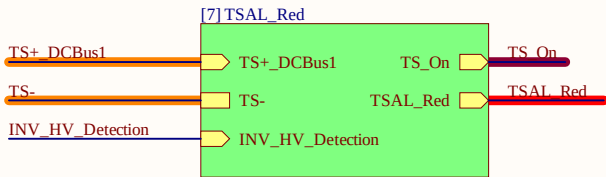
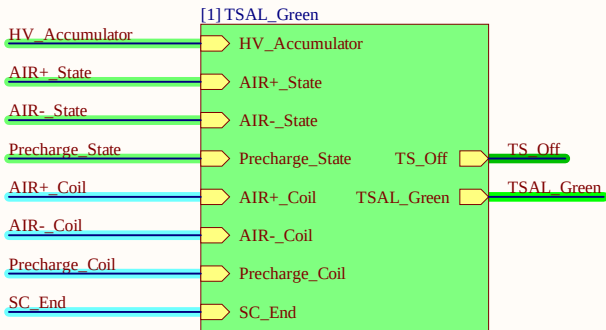
Signal	12 V	1 V	0 V
AIR+ State	AIR+ Open	AIR+ Closed	Broken wire
AIR- State	AIR- Open	AIR- Closed	Broken wire
Precharge State	Precharge Relay Open	Precharge Relay Closed	Broken wire
HV_Accumulator	No HV inside TSAC	HV inside TSAC	Broken wire
Relays_Coil	Relay Open	N/A	Relay Closed



Connectors TS

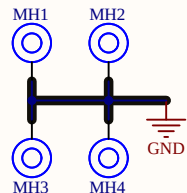


Connectors LV

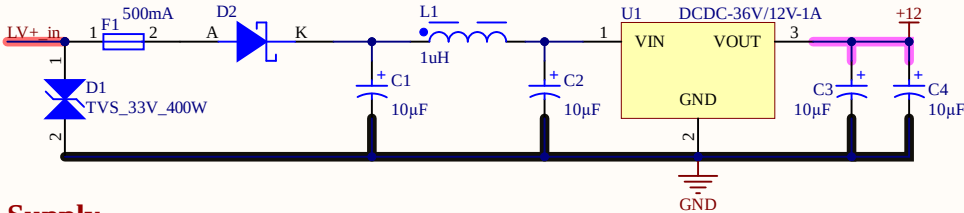
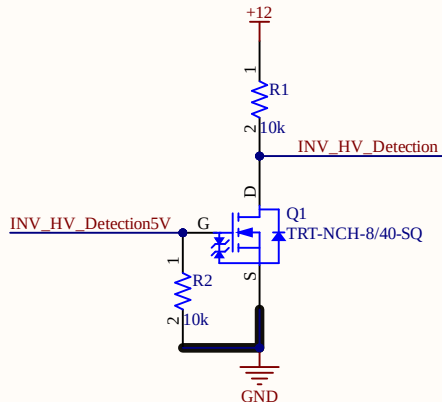


INV_HV_Detection5V is the signal coming from the Inverters that will be implemented in the upcoming years, which transmits a High Logic Level (5 V) when there is <60 V at the DC Bus and a Low Logic Level (0V) when there is >60 V.

Mounting Holes



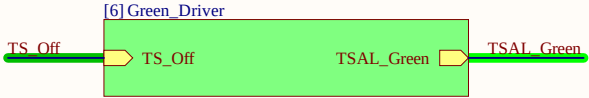
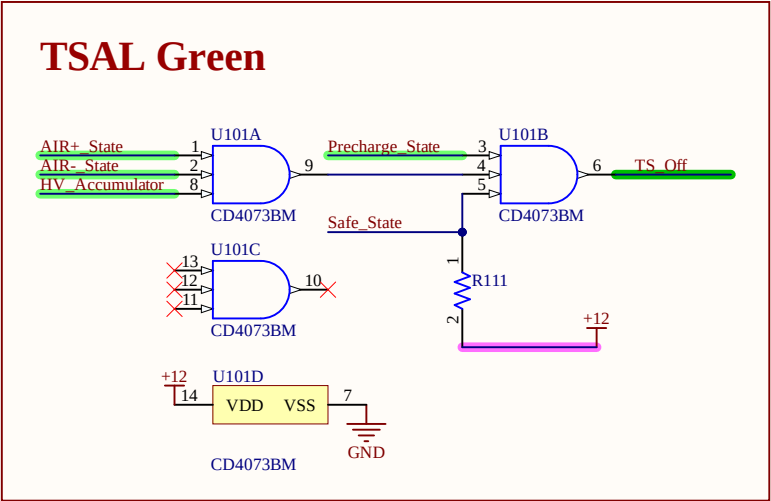
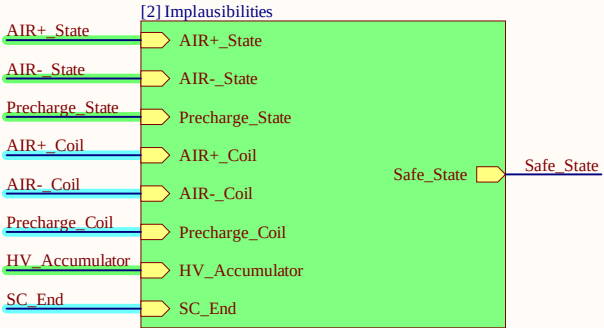
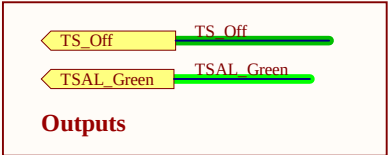
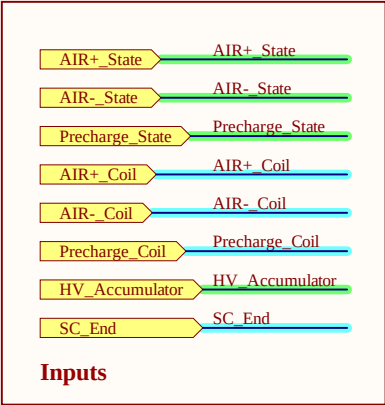
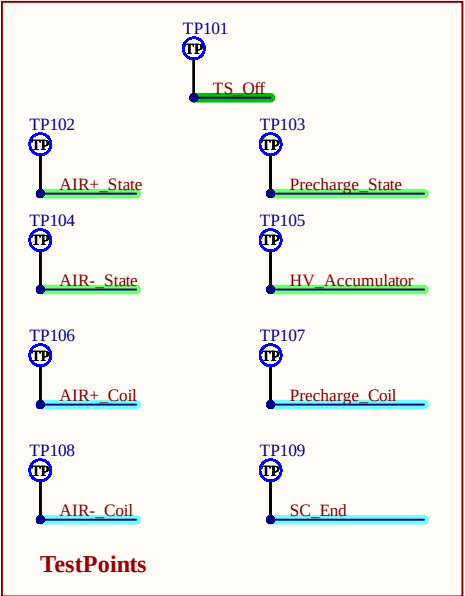
Level Shift



Supply

TSAL CONTROL

Title		
Size	Number	Revision
A4		
Date:	11/11/2024	Sheet of
File:	E:\Documentos\...\TSAL_Control.SchDoc	Drawn By:



Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\bernaO
Project: TSAL Control	Variant: [No Variations]	
Size: -	Page Contents: [1] TSAL_Green.SchDoc	Version:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Department:
Checked by:		Sheet * of *
Title Block_template		Date: 11/11/2024

A

B

C

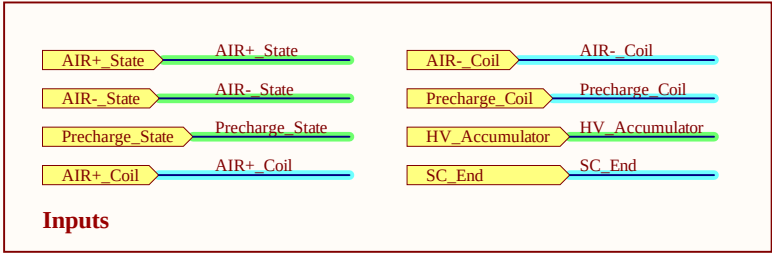
D

A

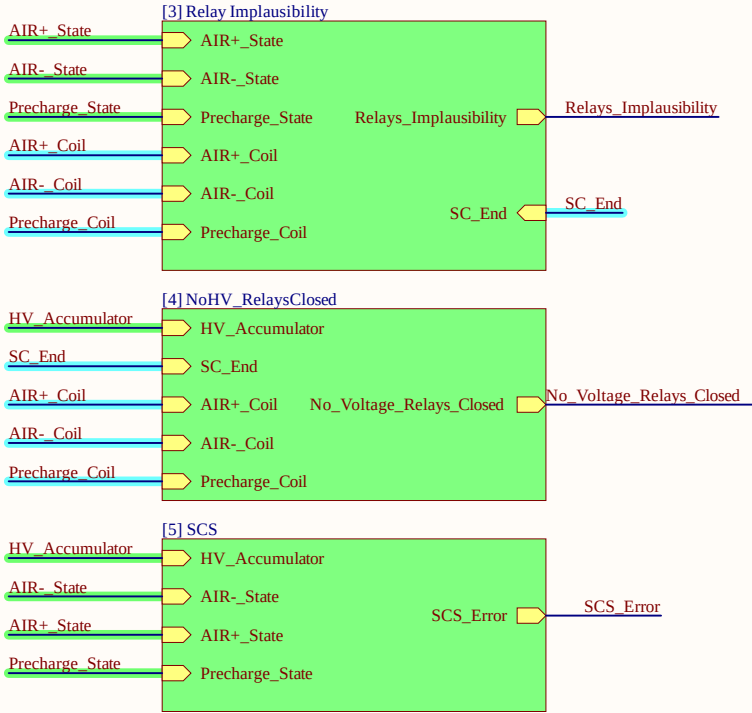
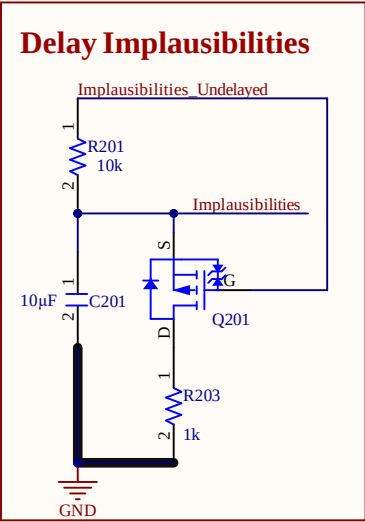
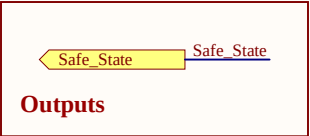
B

C

D



Inputs



HV_Accumulator

HV_Accumulator

SC_End

SC_End

AIR+_Coil

AIR+_Coil

AIR-_Coil

AIR-_Coil

Precharge_Coil

Precharge_Coil

[4] NoHV_RelaysClosed

No_Voltage_Relays_Closed

HV_Accumulator

HV_Accumulator

AIR-_State

AIR-_State

AIR+_State

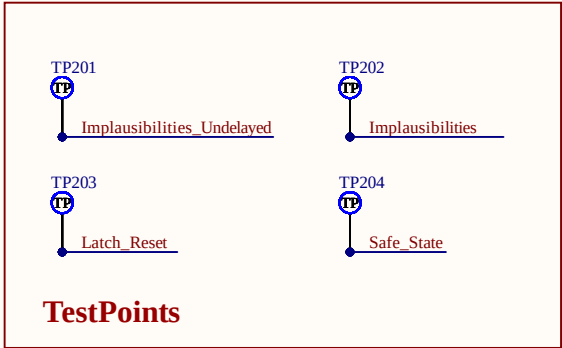
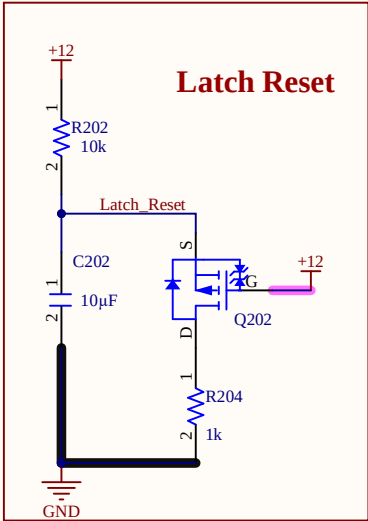
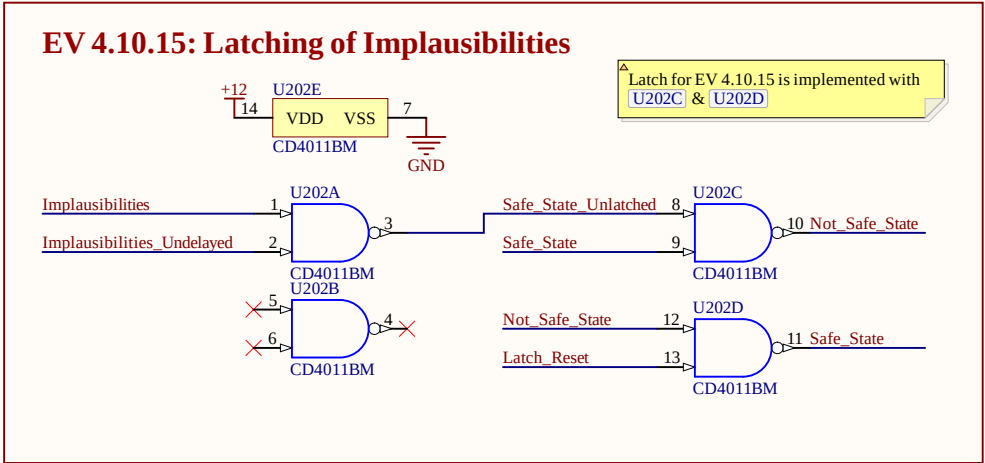
AIR+_State

Precharge_State

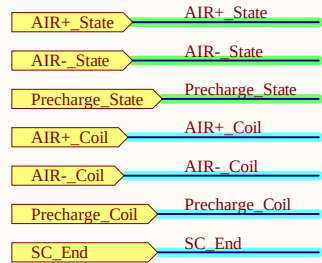
Precharge_State

[5] SCS

SCS_Error



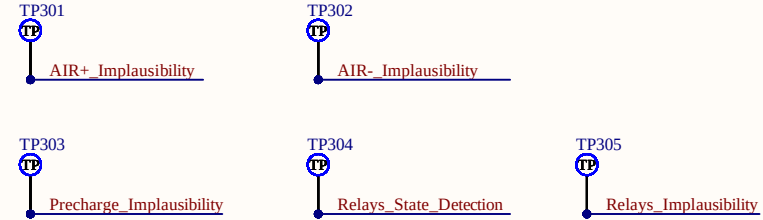
Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\berna\O
Project: TSAL Control Variant: [No Variations]		
Size: -	Page Contents: [2] Implausibilities.SchDoc	Version:
		Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 11/11/2024



Inputs

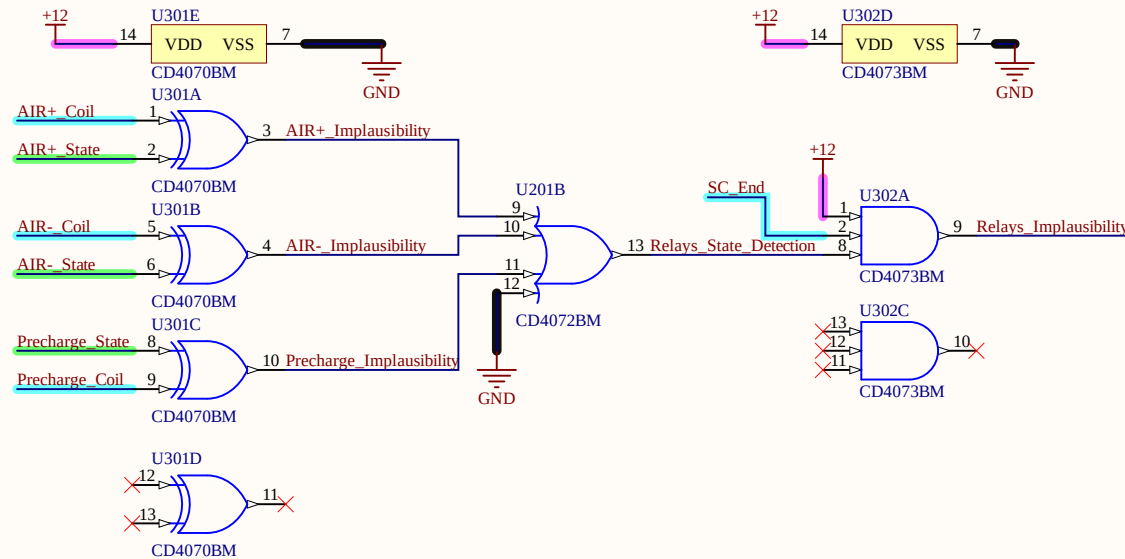
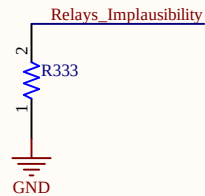
Relays_Implausibility Relays_Implausibility

Outputs



TestPoints

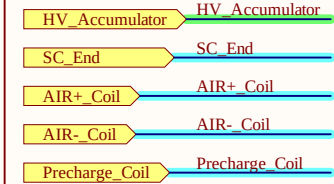
Relay State Implausibility



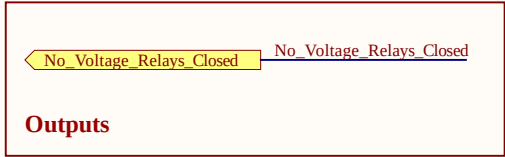
A

A plausibility check against the mechanical state of the relays is implemented on this sheet. [U301A](#) & [U301B](#) & [U301C](#) & [U301D](#) are XOR gates which output a high logic level when the intentional state of any relay (Relay_coil) and the mechanical state (Relay_state) differ. As the Relay's Coil inside the TSAC are supplied directly from the Shutdown Chain, the [U302A](#) conditions the implausibility detection to the SC to be closed.

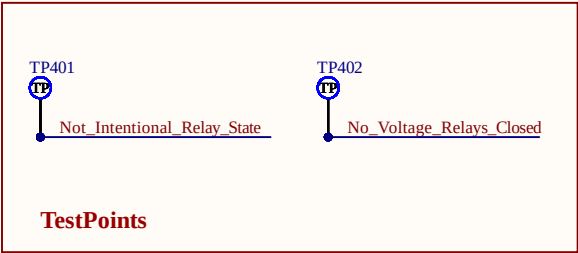
Company: e-Tech Racing		e-techracing.es	Cannot open file C:\Users\berna\O
Project: TSAL Control		Variant: [No Variations]	
Size: -	Page Contents: [3] Relay Implausibility.SchDoc		Version:
			Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com			Sheet * of *
Checked by:			Date: 11/11/2024



Inputs

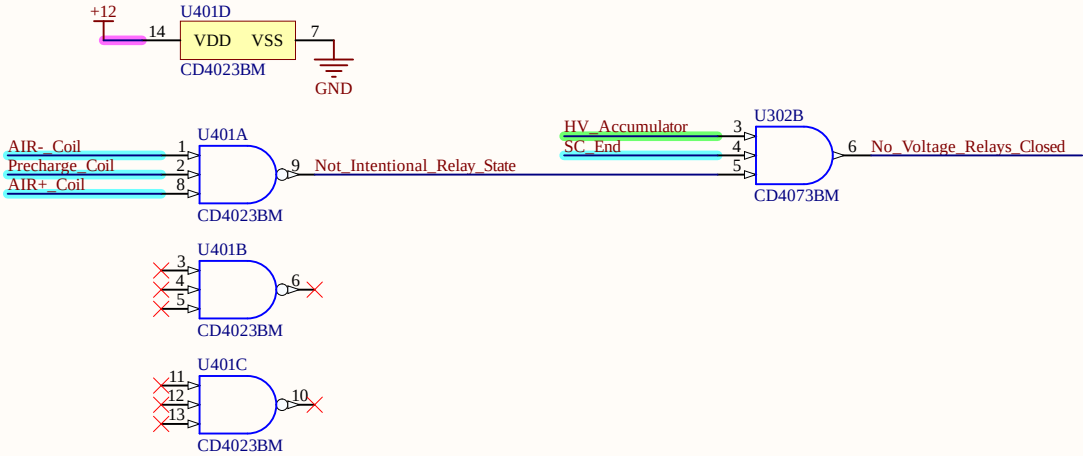


Outputs



TestPoints

EV 4.10.14



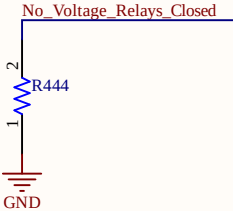
A plausibility check against the intentional state of the relays (AIRs + Precharge) inside the TSAC and the presence of HV inside the Tractive System inside the TSAC is implemented on this sheet.

When the intentional state of any relay is closed (Low Level), the Shutdown Chain is closed and there is no HV inside the TSAC, **Relays_Implausibility** shifts to High Level.

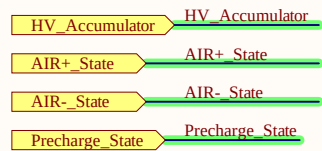
As the intentional state of the Relays is taken from the negative pole of the Relay's coil and all the coils of the Relays influencing the TS inside the TSAC are supplied directly from the SC, **SC_End_30V** coincitions the output of **Relays_Implausibility** in **U302B**.

A The table shown below reflects the Logic implemented in this sheet.

Relay_Coil	HV_Accumulator	Output
L	L	L
L	H	H
H	L	L
H	H	L

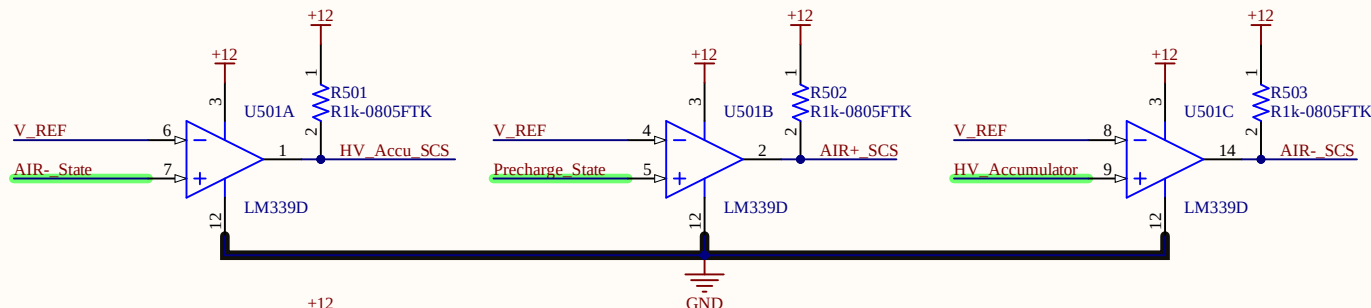
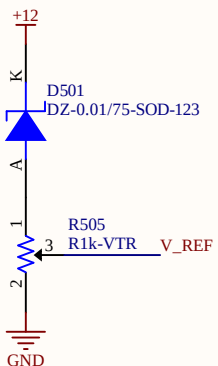
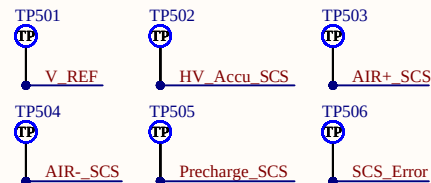


Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\berna\O
Project: TSAL Control Variant: [No Variations]		
Size: -	Page Contents: [4] NoHV_RelaysClosed.SchDoc	Version:
		Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 11/11/2024

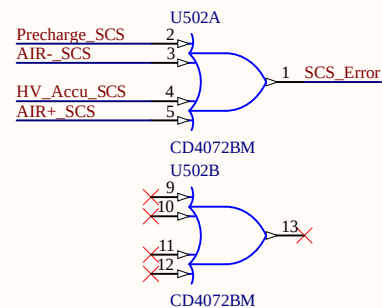


SCS_Error SCS_Error

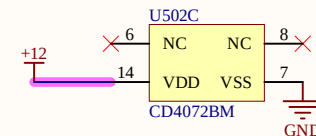
Outputs



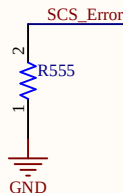
SCS Detection



SCS Error Merge



SCS Failure outputs HIGH LEVEL (12 V) whenever any signal from TSAC is at 0 V. Signals coming from the TSAC regarding the mechanical state of the AIRs and Precharge Relay and the HV Detection inside the Accumulator, follow the configuration shown in the image at the left; meaning whenever any of the mentioned signals arrives at 0 V to this PCB there's a broken wire or it doesn't arrive at the correct node.
This is implemented with U501A & U501B & U501C & U501D.



Resistencia per petarte implausability

Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\bernaO
Project: TSAL Control Variant: [No Variations]		
Size: -	Page Contents: [S]SCS.SchDoc	Version: Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 11/11/2024

A

B

C

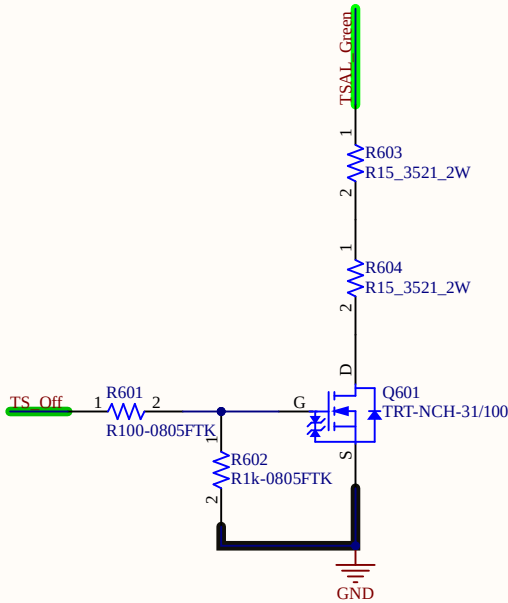
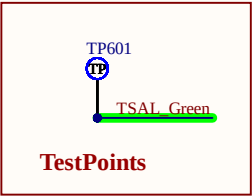
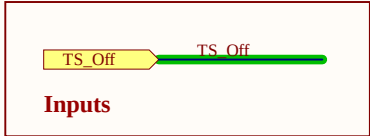
D

A

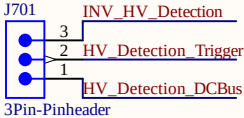
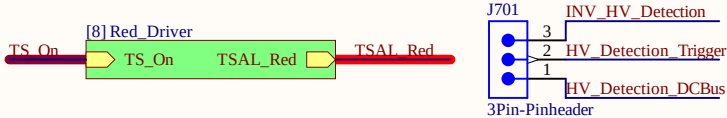
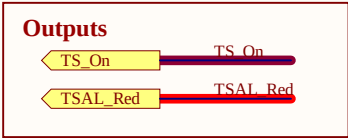
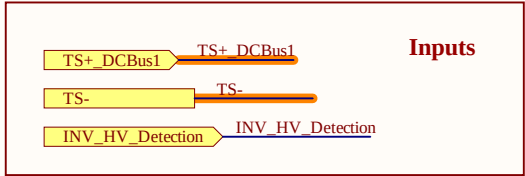
B

C

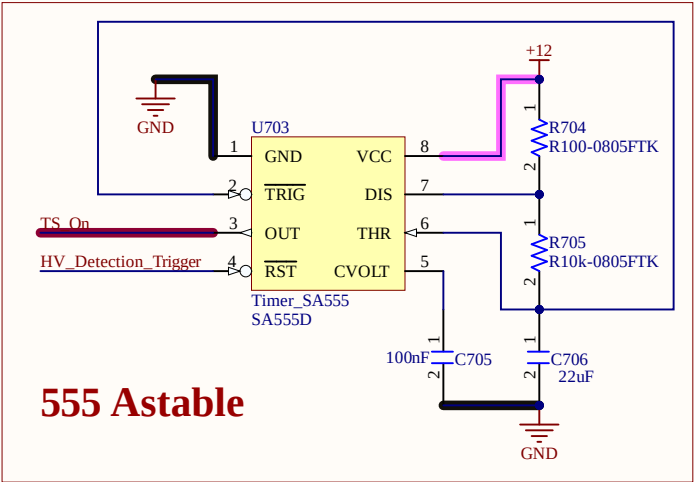
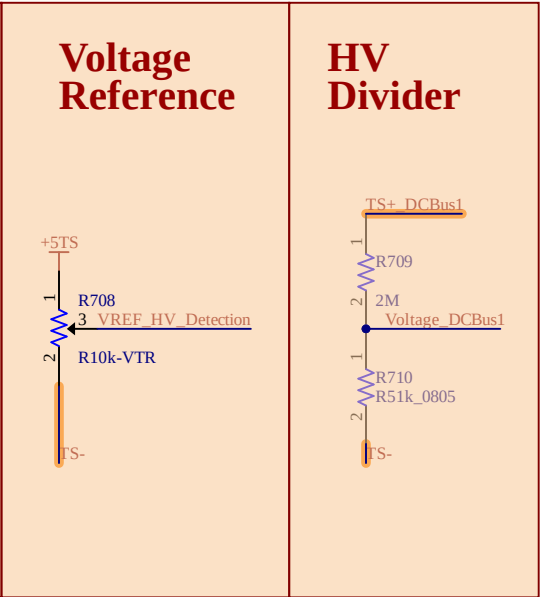
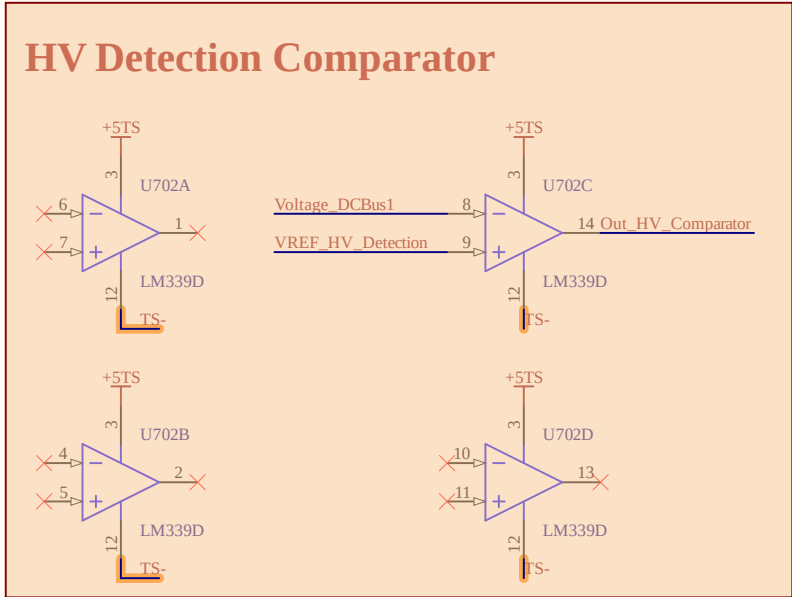
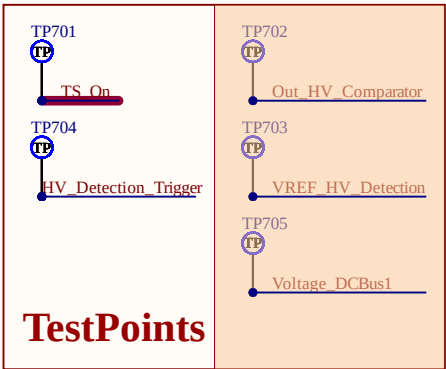
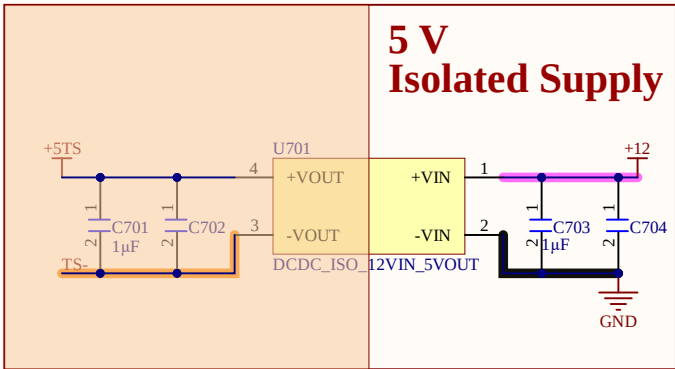
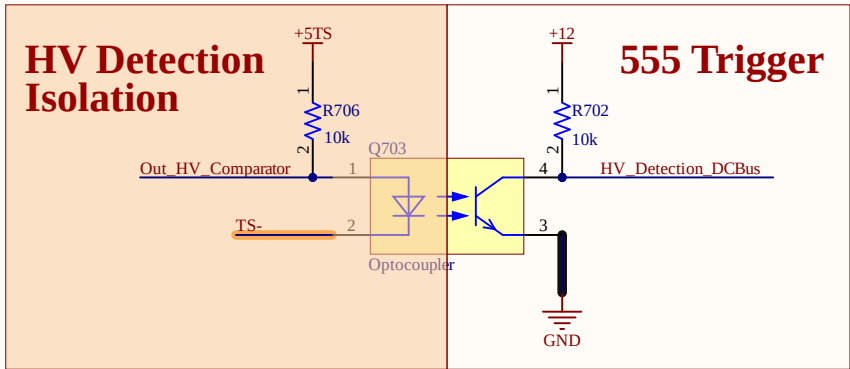
D



Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\bernaO
Project: TSAL Control	Variant: [No Variations]	
Size: -	Page Contents: [6] Green_Driver.SchDoc	Version:
		Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 11/11/2024

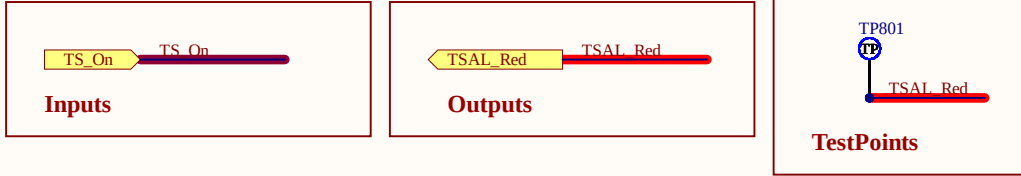


J701 is a 3 pin pinheader that allows selecting the signal that triggers the Timer 555 (U703) to oscillate and drive the TSAL Red.



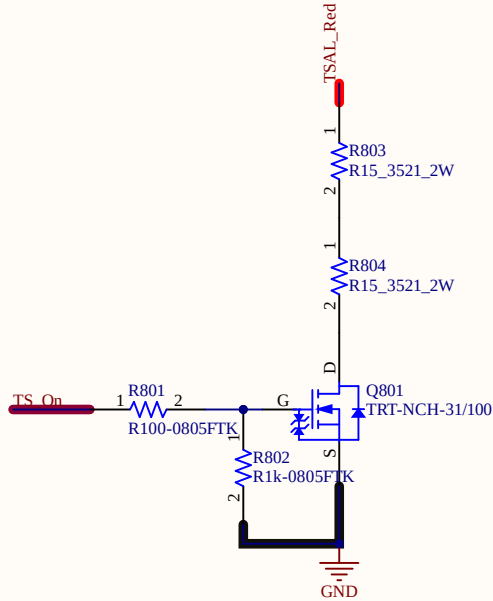
Company: e-Tech Racing e-technacing.es		Cannot open file C:\Users\bernaO
Project: TSAL Control	Variant: [No Variations]	
Size: -	Page Contents: [7] TSAL_Red.SchDoc	Version:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Department:
Checked by:		Sheet * of *
		Date: 11/11/2024

A



A

B



B

C

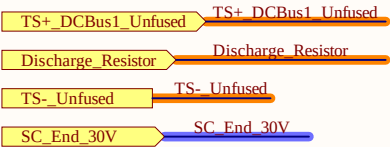
C

D

D

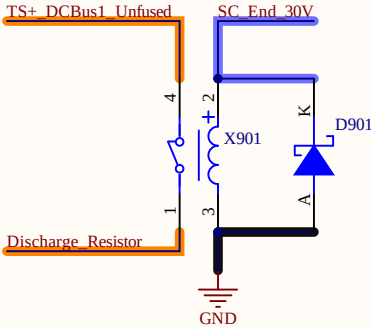
Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\bernaO
Project: TSAL Control	Variant: [No Variations]	
Size: -	Page Contents: [8] Red_Driver.SchDoc	Version:
		Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 11/11/2024

Inputs

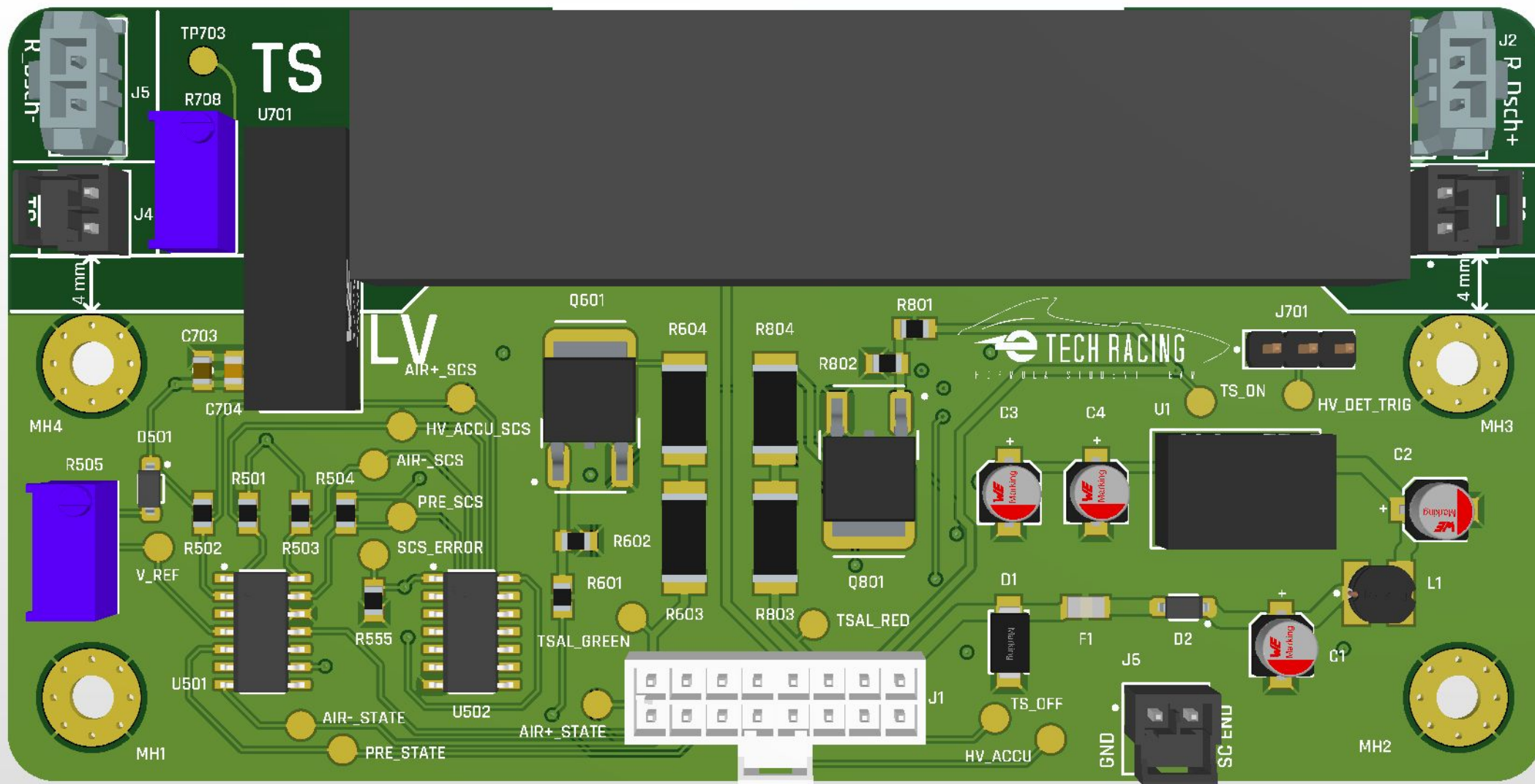


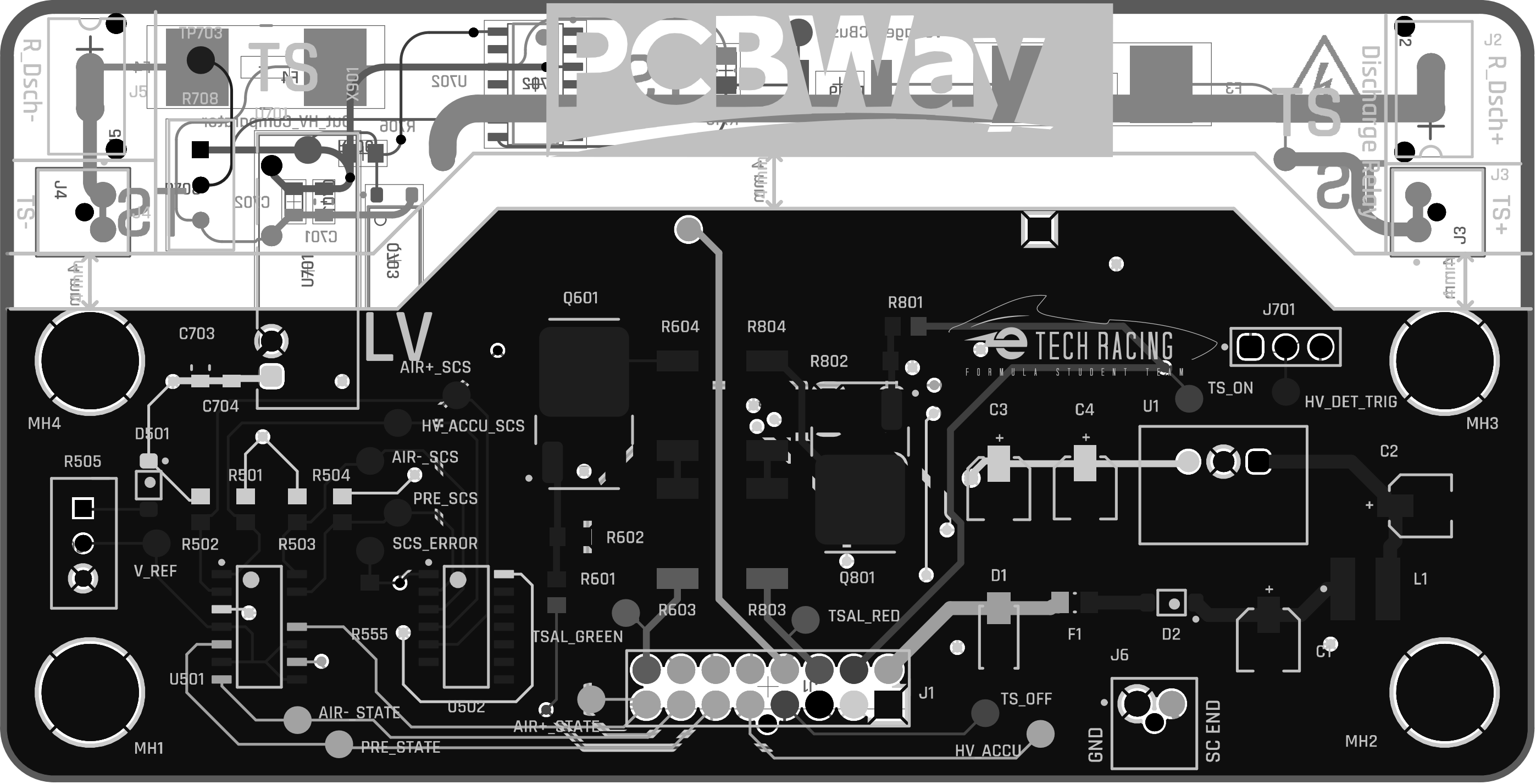
The image at the left, shows the complete discharge circuit implemented. Part of this circuit is implemented on this board, excluding the discharge resistor. The Pull-Up Resistors dissipate 7.2 W in overall at WCE. In order to reduce the thermal stress of the bank resistors, [Not found] & [Not found] & [Not found] are 3W rated and dissipate 1.8 W in WCE each one.

Discharge Relay

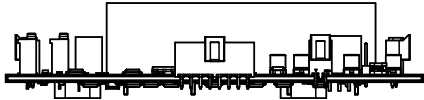


Company: e-Tech Racing e-techracing.es		Cannot open file C:\Users\berna\O
Project: TSAL Control	Variant: [No Variations]	
Size: -	Page Contents: [9] Discharge_Circuit.SchDoc	Version:
Author: Guillem Roperó Serrano guillemroper@gmail.com		Department:
Checked by:		Date: 11/11/2024

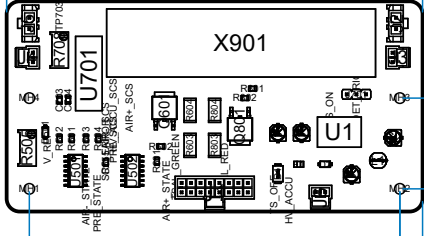




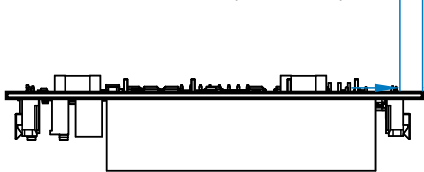
View from Front side (Scale 1:2)



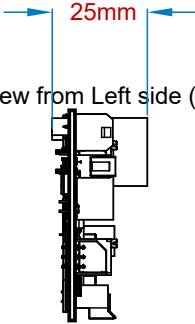
View from Top side (Scale 1:2)



View from Back side (Scale 1:2)



View from Left side (Scale 1:2)



Bill Of Materials

Line #	Designator	Comment	Quantity
1	C1, C2, C3, C4	C10u-35V-Electrolytic	4
2	C201, C202	C10u-16V-0805FNPC-X5R	2
3	C701, C703	C1u-50V-0805FNPC-X7R	2
4	C702, C704, C705	C100n-16V-0805	3
5	C706	C22u-25V-0805	1
6	D1	TVS_33V_400W	1
7	D2, D901	DSC-0.5/30-SOD123	2
8	D501	DZ-0.01/75-SOD-123	1
9	F1	MFU0805FF00500P500	1
10	F3, F4	Fuse_1A_600V	2
11	J1	J_NanoFit_2x8_V	1
12	J2, J5	J_MicroFit_2x1	2
13	J3, J4, J6	J_NanoFit_1x2	3
14	J701	3Pin-Pinheader	1
15	L1	7447730	1
16	MH1, MH2, MH3, MH4	Mounting_Hole_M3	4
17	Q1	TRT-NCH-8/40-SQ	1
18	Q201, Q202	TRT-PCH-2/30-BSS	2
19	Q601, Q801	TRT-NCH-31/100	2
20	Q703	Optocoupler	1
21	R1, R2, R201, R202, R702, R705, R706	R10k-0805FTK	7
22	R3, R5, R7, R9	R12K4-0805-0.125W	4
23	R4, R6, R8, R10	R8.2K_0805_0.125W	4
24	R111, R333, R444, R555	R470-0805FTK	4
25	R203, R204	R1K-0805-0.5W	2
26	R501, R502, R503, R504, R602, R802	R1k-0805FTK	6
27	R505	R1k-VTR	1
28	R601, R704, R801	R100-0805FTK	3
29	R603, R604, R803, R804	R15_3521_2W	4
30	R708	R10k-VTR	1
31	R709	R2M-2512FTK	1
32	R710	R51k_0805	1
33	U1	DCDC-36V/12V-1A	1
34	U101, U302	CD4073BM	2
35	U201, U502	CD4072BM	2
36	U202	CD4011BM	1
37	U301	CD4070BM	1
38	U401	CD4023BM	1
39	U501, U702	LM339D	2
40	U701	DCDC_ISO_12VIN_5VOUT	1
41	U703	Timer_SA555	1
42	X901	HV-Relay-SPST-NC-24V	1